① When can we say that a formula is valid?	$\emptyset \models A$. Where A is a formula of zero-order language.
② 〃 a formula is SAT?	When its single set has a model.
③ 〃 a set of formula is SAT?	When its SAT.
④ 〃 unSAT?	When its not SAT.
⑤ Which symbols are the logical constants of zero-order language?	$\wedge, \vee, \equiv, \supset, \neg$
⑥ Define the sementic equivalence of two formulas.	$A \models B$ and $B \models A$.
⑦ 〃 sementic value of $(A \supset B)$, $(A \wedge B)$, $(A \vee B)$, $(A \equiv B)$.	see table below

A	B	$A \supset B$	$A \wedge B$	$A \vee B$	$A \equiv B$
0	0	1	0	0	1
0	1	1	0	1	0
1	0	0	0	1	0
1	1	1	1	1	1

⑧ 〃 atomic formula.	Non logical constants are atomic formula.
⑨ 〃 interpretation of zero-order language.	$\delta \in \{0,1\}^{Con}$ is the interpretation of zero-order language.
⑩ 〃 CNF.	elementary disjunction.
⑪ 〃 elementary conjunction.	a literal
⑫ 〃 literal.	if $P \in Con$, then P and $\neg P$ formulas are literal.
⑬ Formula labeled the root of construction tree.	atomic formula.

4) $\forall x\,(Q(x,y) \supset \neg(\neg\exists x\,Q(x,z) \supset P(z))) \supset \exists x\,P(x))$

$\forall x_1\,(Q(x_1,y) \supset \neg(\neg\exists x_2\,Q(x_2,z) \supset P(z))) \supset \exists x_3\,P(x_3))$

$\exists_1 \exists_2 \exists_3 \left[(Q(x_1,y) \supset \neg(\neg Q(x_2,z) \supset P(z))) \supset P(x_3)\right]$

5)

C	p	q	$\neg(p \equiv q)$	$((p \supset (q \equiv q)) \supset q)$
X	0	0	0	0 1 1 0 0
	0	1	1	0 1 1 1 1
X	1	0	1	1 1 1 0 0
X	1	1	0	1 1 1 1 1

0 1 1
1 0 0
0 1 1
1 0 0

(6)

	p	q	$(q \equiv p) \supset (p \equiv q)$	$(q \supset \neg q)$
V	0	0	1 1 1	0 1 1
X	0	1	0 1 0	1 0 0
V	1	0	0 1 0	0 1 1
X	1	1	1 1 1	1 0 0

7)

C	p	q	$(q \vee (q \supset (p \supset p)))$	$\neg q \supset q$
X	0	0	1 0 1 1	1 0 0
V	0	1	1 1 1 1	0 1 1
X	1	0	1 0 1 1	1 0 0
V	1	1	1 1 1 1	0 1 1

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$\forall x\,(Q(x,y) \supset \neg(\neg\exists x\,Q(x,z) \supset P(z))) \supset \exists x\,P(x))$

x_1 x_2 x_3

$(\forall x_1\,(Q(x_1,y) \supset \neg(\neg\exists x_2\,Q(x_2,z) \supset P(z))) \supset \exists x_3\,P(x_3))$

$\exists x_1\,\exists x_2\,\exists x_3\,((Q(x_1,y) \supset \neg(\neg Q(x_2,z) \supset P(z))) \supset P(x_3))$

10)

(a) $\exists x_2 \neg P(x_2)$

$P(u_1) = 1$ ¬ | 0
$P(u_2) = 0$ | 1
$P(u_3) = 0$ | 1

(b) $\neg\forall x_2 \neg Q(x_2, x_1)$

0 $Q(u_1, u_2) = 0$ ¬ | 1
0 $Q(u_1, u_3) = 0$ | 1
0 $Q(u_2, u_3) = 0$ | 1

(c) $\forall x_1 \neg P(x_1)$

$P(u_1) = 1$ ¬ | 0
$P(u_2) = 0$ | 1
$P(u_3) = 0$ | 1

d) $\neg\forall x_2\,Q(x_2, x_2)$

¬ 1 | 0 | $Q(u_1, u_1) = 1$
$Q(u_2, u_2) = 0$
$Q(u_3, u_3) = 0$

(e) $\neg\exists x_1\,Q(x_1, x_1)$

¬ 0 | 1 | $Q(u_1, u_1) = 1$
$Q(u_2, u_2) = 0$
$Q(u_3, u_3) = 0$

9) $(r \wedge q \supset \neg p)$

8)

p	q	A: $p \equiv q$	B: $q \wedge q$	C: $(q \wedge q) \wedge p$	$\neg C$
0	0	1	0	0 0 0	1
0	1	0	1	1 0 0	1
1	0	0	0	0 0 1	1
1	1	1	1	1 1 1	0

1 1
0 0
0 1
1 0

a) $\exists x_2 \neg P(x_2)$

1 0 $P(u_1) = 1$ True
1 $P(u_2) = 0$
1 $P(u_3) = 0$

b) $\neg\forall x_2 \neg Q(x_2, x_1)$

0 1 1 $Q(u_1, u_2) = 0$ False
1 $Q(u_1, u_3) = 0$
1 $Q(u_2, u_3) = 0$

c) $\forall x_1 \neg P(x_1)$

0 0 $P(u_1) = 1$ False
1 $P(u_2) = 0$
1 $P(u_3) = 0$

d) $\neg\forall x_2\,Q(x_2, x_2)$

1 0 $Q(u_1, u_1) = 1$ True
$Q(u_2, u_2) = 0$
$Q(u_3, u_3) = 0$

e) $\neg\exists x_1\,Q(x_1, x_1)$

0 1 $Q(u_1, u_1) = 1$ False
$Q(u_2, u_2) = 0$
$Q(u_3, u_3) = 0$

$\emptyset$

$\rightarrow \equiv \quad M, A \equiv B \rightarrow A, M, B, \quad B, M, A$

$\rightarrow \supset \quad M, A \supset B \rightarrow A, M, B$

$\supset \rightarrow \quad A \supset B, M \rightarrow M, A; \; B; M$

$\rightarrow \wedge \quad M, A \wedge B \rightarrow M, A \; ; \; M, B$

$\wedge \rightarrow \quad A \wedge B, M \rightarrow A, B, M$

$\rightarrow \vee \quad M, A \vee B \rightarrow M, A, B$

$\vee \rightarrow \quad A \vee B, M \rightarrow \cancel{A, B, M} \quad A, M \quad B, M$

$\rightarrow q, p, \neg(q \wedge r); \; p \rightarrow \neg(q \wedge r)$

$\underbrace{\neg(q \wedge r)}_{A} \supset \underbrace{p}_{B} \rightarrow q, p$

$\cancel{\neg(q \wedge r) \supset p, q, p}$

$$\frac{\neg(q \wedge r) \supset p \rightarrow (q \vee p); \quad \neg(q \wedge r) \supset p \rightarrow (\neg p \supset r)}{\underbrace{\neg(q \wedge r) \supset p}_{M} \vDash (\underbrace{(q \vee p)}_{A} \wedge \underbrace{(\neg p \supset r)}_{B})}$$

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$\neg(q \wedge r) \supset p \;\; \vDash \;\; (q \vee p) \wedge (\neg p \supset r)$

$$\frac{\dfrac{\dfrac{\dfrac{\overset{ax.}{q, r \rightarrow q, p}}{q, r \rightarrow (q \vee p)}}{(q \wedge r) \rightarrow (q \vee p)} \qquad \overset{ax.}{p \rightarrow q, p}}{\rightarrow (q \vee p), \neg(q \wedge r) \; ; \; p \rightarrow (q \vee p)}\,(\rightarrow\vee) \qquad \dfrac{\dfrac{\dfrac{\dfrac{\overset{ax.}{q, r \rightarrow p, r}}{(q \wedge r) \rightarrow p, r}}{\neg p, (q \wedge r) \rightarrow r}}{(q \wedge r) \rightarrow (\neg p \supset r)}\,(\rightarrow\supset) \qquad \dfrac{\dfrac{\overset{ax.}{p \rightarrow p, r}}{\neg p, p \rightarrow r}}{p \rightarrow (\neg p \supset r)}}{\rightarrow (\neg p \supset r), \neg(q \wedge r) \; ; \; p \rightarrow (\neg p \supset r)}\,(\rightarrow\supset)}{\dfrac{\neg(q \wedge r) \supset p \rightarrow (q \vee p) \quad ; \quad \neg(q \wedge r) \supset p \rightarrow (\neg p \supset r)}{\neg(q \wedge r) \supset p \longrightarrow (q \vee p) \wedge (\neg p \supset r)}\,(\rightarrow\wedge)}\,(\supset\rightarrow)$$

$$(\rightarrow \equiv)\;\frac{A, \Gamma \rightarrow \Delta, B \;\; ; \;\; B, \Gamma \rightarrow \Delta, A}{\Gamma \rightarrow \Delta, (A \equiv B)} \qquad (\equiv \rightarrow)\;\frac{A, B, \Gamma \rightarrow \Delta \;\; ; \;\; \Gamma \rightarrow \Delta, A, B}{(A \equiv B), \Gamma \rightarrow \Delta}$$

$$(\rightarrow \supset)\;\frac{A, \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \supset B)} \qquad (\supset \rightarrow)\;\frac{\Gamma \rightarrow \Delta, A \; ; \; B, \Gamma \rightarrow \Delta}{(A \supset B), \Gamma \rightarrow \Delta}$$

$$(\rightarrow \wedge)\;\frac{\Gamma \rightarrow \Delta, A \;\; ; \;\; \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \wedge B)} \qquad (\wedge \rightarrow)\;\frac{A, B, \Gamma \rightarrow \Delta}{(A \wedge B), \Gamma \rightarrow \Delta}$$

$$(\rightarrow \vee)\;\frac{\Gamma \rightarrow \Delta, A, B}{\Gamma \rightarrow \Delta, (A \vee B)} \qquad (\vee \rightarrow)\;\frac{A, \Gamma \rightarrow \Delta \; ; \; B, \Gamma \rightarrow \Delta}{(A \vee B), \Gamma \rightarrow \Delta}$$

3.) Prove the following with sequent calculus or natural deduction.

$$\neg(q \wedge r) \supset p \vDash ((p \vee q) \wedge (\neg p \supset r))$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\overset{ax}{q, r \to p, q}}{q, r \to (p \vee q)}(\to\vee) \;;\; \dfrac{\dfrac{\overset{ax}{q, r \to r, p}}{\neg p, q, r \to r}(\neg\to)}{q, r \to (\neg p \supset r)}(\to\supset)}{q, r \to ((p \vee q) \wedge (\neg p \supset r))}(\to\wedge)}{(q \wedge r) \to ((p \vee q) \wedge (\neg p \supset r))}(\wedge\to)}{\to ((p \vee q) \wedge (\neg p \supset r)), \neg(q \wedge r)}(\to\neg) \;;\; \dfrac{\dfrac{\overset{ax}{p \to p, q}}{p \to (p \vee q)}(\to\vee) \;;\; \dfrac{\dfrac{\overset{ax}{p \to r, p}}{\neg p, p \to r}(\neg\to)}{p \to (\neg p \supset r)}(\to\supset)}{p \to ((p \vee q) \wedge (\neg p \supset r))}(\to\wedge)}{\neg(q \wedge r) \supset p \to ((p \vee q) \wedge (\neg p \supset r))}(\supset\to)$$

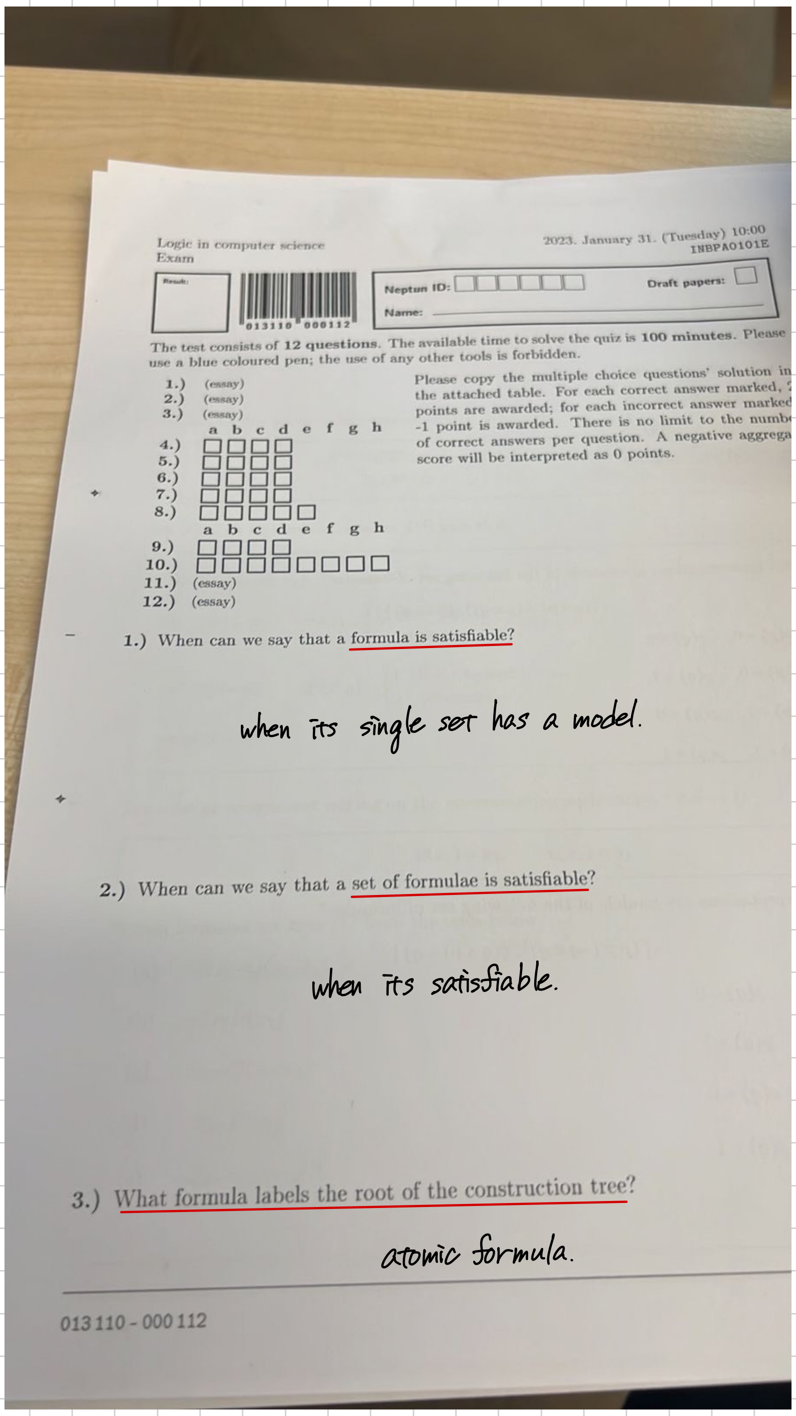
$$(\to\equiv)\ \dfrac{A, \Gamma \to \Delta, B \;;\; B, \Gamma \to \Delta, A}{\Gamma \to \Delta, (A \equiv B)} \qquad (\equiv\to)\ \dfrac{A, B, \Gamma \to \Delta \;;\; \Gamma \to \Delta, A, B}{(A \equiv B), \Gamma \to \Delta}$$

$$(\to\supset)\ \dfrac{A, \Gamma \to \Delta, B}{\Gamma \to \Delta, (A \supset B)} \qquad (\supset\to)\ \dfrac{\Gamma \to \Delta, A \;;\; B, \Gamma \to \Delta}{(A \supset B), \Gamma \to \Delta}$$

$$(\to\wedge)\ \dfrac{\Gamma \to \Delta, A \;;\; \Gamma \to \Delta, B}{\Gamma \to \Delta, (A \wedge B)} \qquad (\wedge\to)\ \dfrac{A, B, \Gamma \to \Delta}{(A \wedge B), \Gamma \to \Delta}$$

$$(\to\vee)\ \dfrac{\Gamma \to \Delta, A, B}{\Gamma \to \Delta, (A \vee B)} \qquad (\vee\to)\ \dfrac{A, \Gamma \to \Delta \;;\; B, \Gamma \to \Delta}{(A \vee B), \Gamma \to \Delta}$$

$$\dfrac{\dfrac{\dfrac{\overset{ax.}{q, r \to p, q}}{(q \wedge r) \to (p \vee q)} \;;\; \dfrac{\dfrac{\dfrac{\overset{ax.}{q, r \to p, r}}{\neg p, q, r \to r}}{q, r \to \neg p \supset r}}{(q \wedge r) \to (\neg p \supset r)}}{\dfrac{(q \wedge r) \to (p \vee q) \wedge (\neg p \supset r)}{\to (p \vee q) \wedge (\neg p \supset r), \neg(q \wedge r)}} \;;\; \dfrac{\dfrac{\overset{ax.}{p \to p, q}}{p \to (p \vee q)} \;;\; \dfrac{\dfrac{\overset{ax.}{p \to p, r}}{\neg p, p \to r}}{p \to (\neg p \supset r)}}{p \to (p \vee q) \wedge (\neg p \supset r)}}{\neg(q \wedge r) \supset p \to (p \vee q) \wedge (\neg p \supset r)}$$



Logic in computer science
Exam

2023. January 31. (Tuesday) 10:00
INBPA0101E

Result:

Neptun ID: | Draft papers:

Name:

The test consists of **12 questions**. The available time to solve the quiz is **100 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) (essay)
2.) (essay)
3.) (essay)
4.)–8.) a b c d e f g h
9.)–10.) a b c d e f g h
11.) (essay)
12.) (essay)

1.) When can we say that a formula is satisfiable?

when its single set has a model.

2.) When can we say that a set of formulae is satisfiable?

when its satisfiable.

3.) What formula labels the root of the construction tree?

atomic formula.

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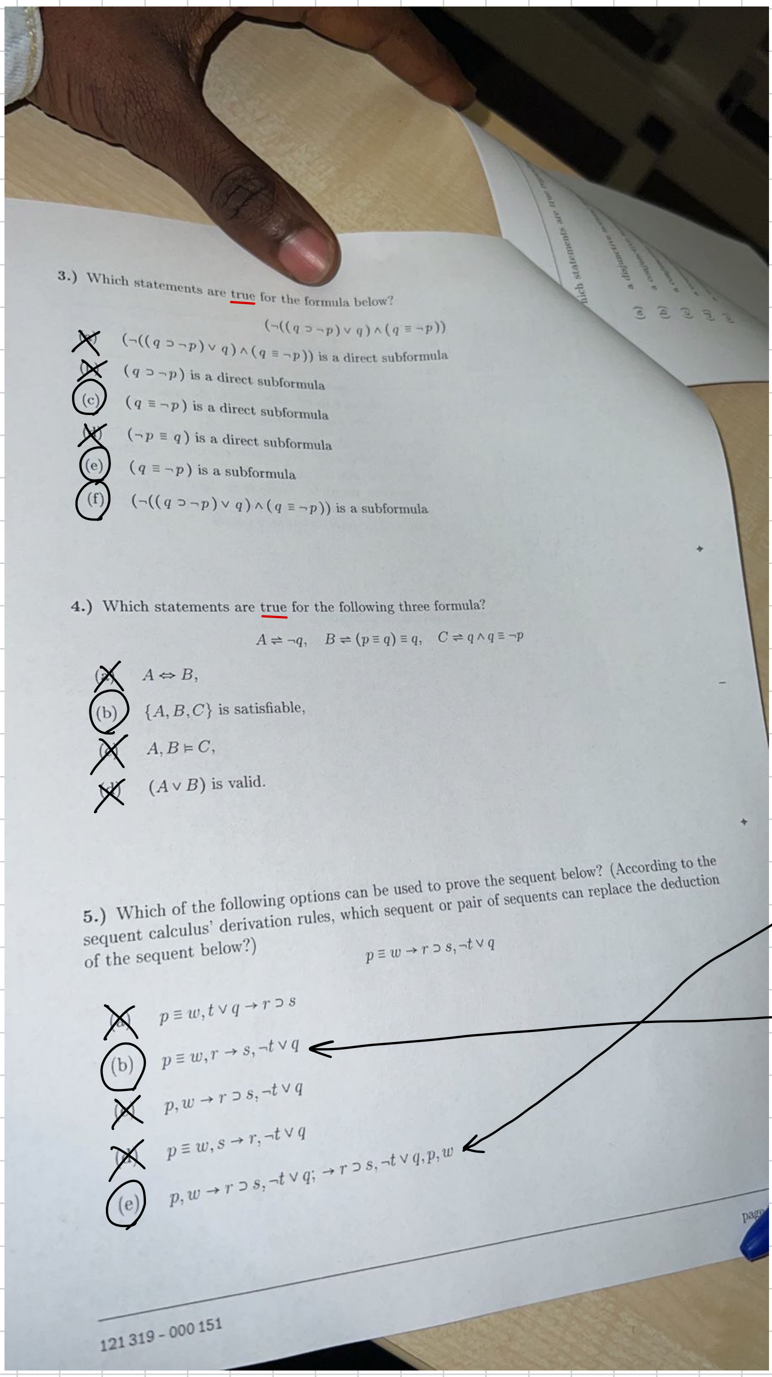

3.) Which statements are true for the formula below?

$(\neg((q \supset \neg p) \vee q) \wedge (q \equiv \neg p))$

(a) $(\neg((q \supset \neg p) \vee q) \wedge (q \equiv \neg p))$ is a direct subformula
(b) $(q \supset \neg p)$ is a direct subformula
(c) $(q \equiv \neg p)$ is a direct subformula
(d) $(\neg p \equiv q)$ is a direct subformula
(e) $(q \equiv \neg p)$ is a subformula
(f) $(\neg((q \supset \neg p) \vee q) \wedge (q \equiv \neg p))$ is a subformula

4.) Which statements are true for the following three formula?

$A \rightleftharpoons \neg q, \quad B \rightleftharpoons (p \equiv q) \equiv q, \quad C \rightleftharpoons q \wedge q \equiv \neg p$

(a) $A \Leftrightarrow B$,
(b) $\{A, B, C\}$ is satisfiable,
(c) $A, B \vDash C$,
(d) $(A \vee B)$ is valid.

5.) Which of the following options can be used to prove the sequent below? (According to the sequent calculus' derivation rules, which sequent or pair of sequents can replace the deduction of the sequent below?)

$p \equiv w \to r \supset s, \neg t \vee q$

(a) $p \equiv w, t \vee q \to r \supset s$
(b) $p \equiv w, r \to s, \neg t \vee q$
(c) $p, w \to r \supset s, \neg t \vee q$
(d) $p \equiv w, s \to r, \neg t \vee q$
(e) $p, w \to r \supset s, \neg t \vee q; \to r \supset s, \neg t \vee q, p, w$

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③ ① $(\neg((q \supset \neg p) \vee q) \wedge (q \equiv \neg p))$

② $\neg((q \supset \neg p) \vee q)$ ③ $(q \equiv \neg p)$ direct subformula + subformula

④ $((q \supset \neg p) \vee q)$ q $\neg p$

⑤ $(q \supset \neg p)$ q ダブり p

⑥ q ⑦ $\neg p$

⑧ p 8 subformulas

④ a) $\cancel{A \Leftrightarrow B}$: $A \vDash B$, $B \vDash A$

$\cancel{A \vDash B}$ has model → $\cancel{SAT}$

p	q	¬q	¬((p≡q)≡q)		
0	0	1	1	1	0
0	1	0	1	0	0
1	0	1	0	0	1
1	1	0	0	1	1

b)

A	B	(q∧q)≡¬p		
1	0	0	0	1
0	0	1	1	1
1	1	0	1	0
0	1	1	0	0

c)

A	B	¬C
1	0	1
0	0	0
1	1	0
0	1	1

it has no model → unSAT

d) A B

$1 \vee 0 = 1$
$0 \vee 0 = 0$
$1 \vee 1 = 1$
$0 \vee 1 = 1$

0があるから valid じゃない

⑤

$(\to \equiv)\ \dfrac{A, \Gamma \to \Delta, B \ ;\ B, \Gamma \to \Delta, A}{\Gamma \to \Delta, (A \equiv B)}$ $(\equiv \to)\ \dfrac{A, B, \Gamma \to \Delta \ ;\ \Gamma \to \Delta, A, B}{(A \equiv B), \Gamma \to \Delta}$

$(\to \supset)\ \dfrac{A, \Gamma \to \Delta, B}{\Gamma \to \Delta, (A \supset B)}$ $(\supset \to)\ \dfrac{\Gamma \to \Delta, A \ ;\ B, \Gamma \to \Delta}{(A \supset B), \Gamma \to \Delta}$

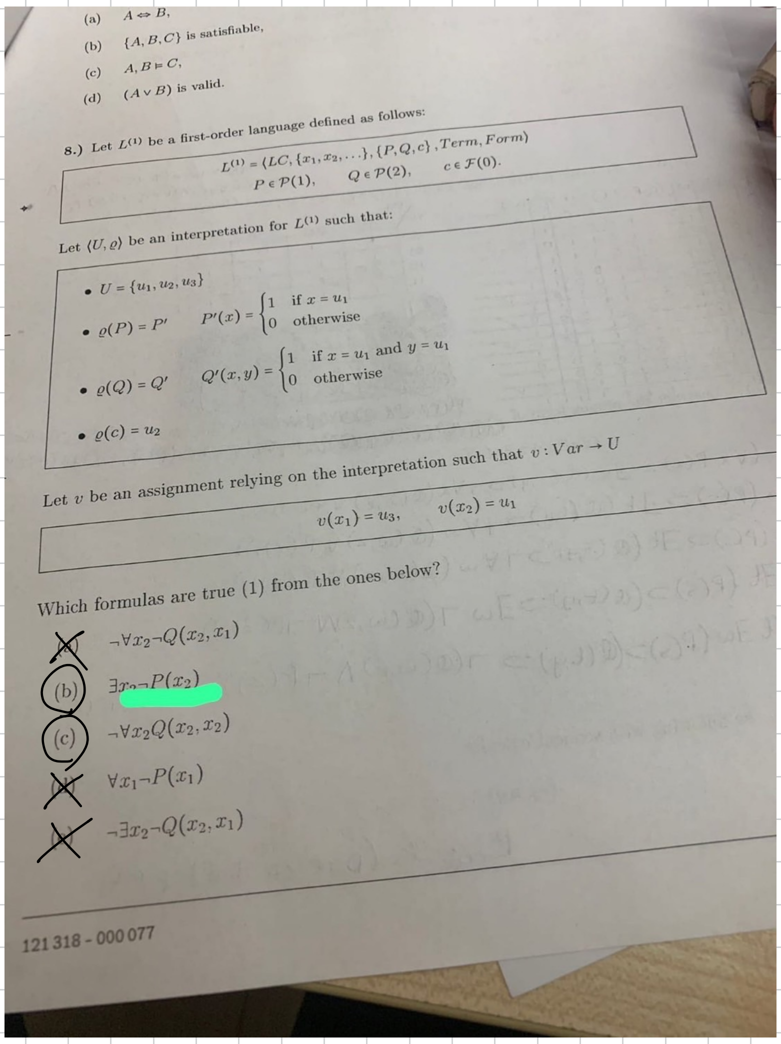
$(\to \wedge)\ \dfrac{\Gamma \to \Delta, A \ ;\ \Gamma \to \Delta, B}{\Gamma \to \Delta, (A \wedge B)}$ $(\wedge \to)\ \dfrac{A, B, \Gamma \to \Delta}{(A \wedge B), \Gamma \to \Delta}$

$(\to \vee)\ \dfrac{\Gamma \to \Delta, A, B}{\Gamma \to \Delta, (A \vee B)}$ $(\vee \to)\ \dfrac{A, \Gamma \to \Delta \ ;\ B, \Gamma \to \Delta}{(A \vee B), \Gamma \to \Delta}$

① $\dfrac{p, w \to r \supset s, \neg t \vee q \ ;\ \to r \supset s, \neg t \vee q, p, w}{p \equiv w \to r \supset s, \neg t \vee q}$ $(\equiv \to)$

② $\dfrac{r, p \equiv w \to \neg t \vee q, s}{p \equiv w \to r \supset s, \neg t \vee q}$ $(\to \supset)$

③ $\dfrac{p \equiv w \to r \supset s, \neg t, q}{p \equiv w \to r \supset s, \neg t \vee q}$ $(\to \vee)$



(a) $A \Leftrightarrow B$,
(b) $\{A, B, C\}$ is satisfiable,
(c) $A, B \vDash C$,
(d) $(A \vee B)$ is valid.

8.) Let $L^{(1)}$ be a first-order language defined as follows:

$$L^{(1)} = (LC, \{x_1, x_2, \dots\}, \{P, Q, c\}, Term, Form)$$
$$P \in \mathcal{P}(1), \quad Q \in \mathcal{P}(2), \quad c \in \mathcal{F}(0).$$

Let (U, ϱ) be an interpretation for $L^{(1)}$ such that:

- $U = \{u_1, u_2, u_3\}$
- $\varrho(P) = P'$ $\quad P'(x) = \begin{cases} 1 & \text{if } x = u_1 \\ 0 & \text{otherwise} \end{cases}$
- $\varrho(Q) = Q'$ $\quad Q'(x,y) = \begin{cases} 1 & \text{if } x = u_1 \text{ and } y = u_1 \\ 0 & \text{otherwise} \end{cases}$
- $\varrho(c) = u_2$

Let v be an assignment relying on the interpretation such that $v : Var \to U$

$$v(x_1) = u_3, \quad v(x_2) = u_1$$

Which formulas are true (1) from the ones below?

(a) $\neg\forall x_2 \neg Q(x_2, x_1)$
(b) $\exists x_2 \neg P(x_2)$
(c) $\neg\forall x_2 Q(x_2, x_2)$
(d) $\forall x_1 \neg P(x_1)$
(e) $\neg\exists x_2 \neg Q(x_2, x_1)$

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⑧ a) $\neg\forall x_2 \neg Q(x_2, x_1)$

$0 \quad 1 \quad 1 \; Q(u_1, u_2) = 0$ False

$1 \; Q(u_1, u_3) = 0$

$1 \; Q(u_2, u_3) = 0$

b) $\exists x_2 \neg P(x_2)$

$1 \quad 0 \; P(u_1) = 1$ True

$1 \; P(u_2) = 0$

$1 \; P(u_3) = 0$

c) $\neg\forall x_2 Q(x_2, x_2)$

$1 \quad 0 \; Q(u_1, u_1) \; 1$ True

$Q(u_2, u_2) \; 0$

$Q(u_3, u_3) \; 0$

d) $\forall x_1 \neg P(x_1)$

$0 \quad 0 \; P(u_1) = 1$ False

$1 \; P(u_2) = 0$

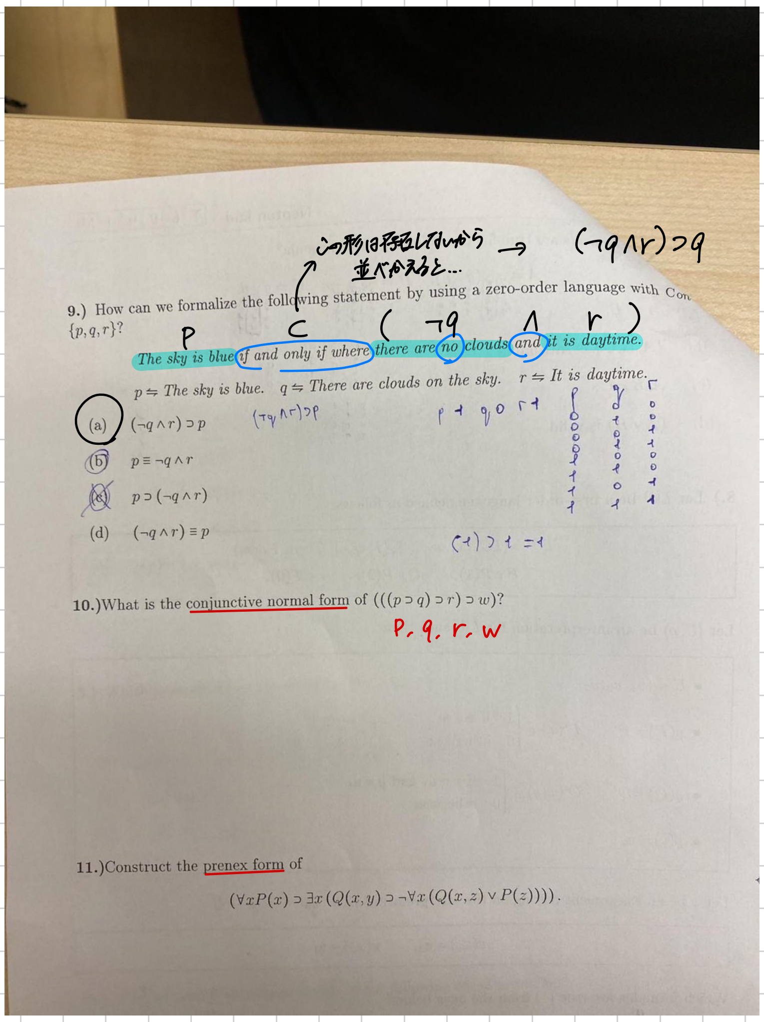
$1 \; P(u_3) = 0$

e) $\neg\exists x_2 \neg Q(x_2, x_1)$

$0 \quad 1 \quad 1 \; Q(u_1, u_2) = 0$ False

$1 \; Q(u_1, u_3) = 0$

$1 \; Q(u_2, u_3) = 0$



この形は存在しないから → $(\neg q \wedge r) \supset q$
並べかえると…

9.) How can we formalize the following statement by using a zero-order language with Con $\{p, q, r\}$?

The sky is blue if and only if where there are no clouds and it is daytime.

p ⇋ *The sky is blue.* $\quad q$ ⇋ *There are clouds on the sky.* $\quad r$ ⇋ *It is daytime.*

(a) $(\neg q \wedge r) \supset p$
(b) $p \equiv \neg q \wedge r$
(c) $p \supset (\neg q \wedge r)$
(d) $(\neg q \wedge r) \equiv p$

10.) What is the conjunctive normal form of $(((p \supset q) \supset r) \supset w)$?

p, q, r, w

11.) Construct the prenex form of

$$(\forall x P(x) \supset \exists x(Q(x,y) \supset \neg\forall x(Q(x,z) \vee P(z)))).$$

⑩

① $\neg\neg A \Leftrightarrow A$
② $A \supset B \Leftrightarrow \neg A \vee B$
③ $\neg(A \wedge B) \Leftrightarrow \neg A \vee \neg B$
④ $A \wedge (B \vee C) \Leftrightarrow (A \wedge B) \vee (A \wedge C)$

$((p \supset q) \supset r) \supset w$

$((\neg p \vee q) \supset r) \supset w$

$(\neg(\neg p \vee q) \vee r) \supset w$

$\neg(\neg(\neg p \vee q) \vee r) \vee w$

$\neg((p \wedge \neg q) \vee r) \vee w$

$((\neg p \vee q) \wedge \neg r) \vee w$

$((\neg r \wedge \neg p) \vee (\neg r \wedge q)) \vee w$ ← DNF

$(\neg r \vee w) \wedge ((\neg p \vee q) \vee w)$ ← CNF

$((p \supset q) \supset r) \supset w$

$((\neg p \vee q) \supset r) \supset w$

$(\neg(\neg p \vee q) \vee r) \supset w$

$\neg(\neg(\neg p \vee q) \vee r) \vee w$

$\neg((p \wedge \neg q) \vee r) \vee w$

$((\neg p \vee q) \wedge \neg r) \vee w$

$(\neg r \wedge \neg p) \vee (\neg r \wedge q) \vee w$ ← DNF

$(\neg r \vee w) \wedge ((\neg p \vee q) \vee w)$ ← CNF

⑪ $(\forall \underbrace{x}_{x_1} P(x) \supset \exists \underbrace{x}_{x_2} (Q(x,y) \supset \neg\forall \underbrace{x}_{x_3} (Q(x,z) \vee P(z))))$

$(\forall x_1 P(x_1) \supset \exists x_2 (Q(x_2, y) \supset \neg\forall x_3 (Q(x_3, z) \vee P(z))))$

$\exists x_1 \exists x_2 \exists x_3 (P(x_1) \supset (Q(x_2, y) \supset \neg(Q(x_3, z) \vee P(z))))$

1.) Define the semantic equivalence of two formulas!

Score: 2 / 2.

$A \models B$ and $B \models A$.

2.) When can we say that a set of formulae is satisfiable?

Score: 2 / 2.

when its satisfiable.

3.) Prove the following with sequent calculus or natural deduction.

$$\neg(q \wedge r) \supset p \models ((p \vee q) \wedge (\neg p \supset r))$$

Score: 2 / 2.

③

$(\rightarrow\equiv)\ \dfrac{A, \Gamma \rightarrow \Delta, B\ ;\ B, \Gamma \rightarrow \Delta, A}{\Gamma \rightarrow \Delta, (A \equiv B)}$ $\qquad (\equiv\rightarrow)\ \dfrac{A, B, \Gamma \rightarrow \Delta\ ;\ \Gamma \rightarrow \Delta, A, B}{(A \equiv B)\ \Gamma \rightarrow \Delta}$

$(\rightarrow\supset)\ \dfrac{A, \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \supset B)}$ $\qquad (\supset\rightarrow)\ \dfrac{\Gamma \rightarrow \Delta, A\ ;\ B, \Gamma \rightarrow \Delta}{(A \supset B)\ \Gamma \rightarrow \Delta}$

$(\rightarrow\wedge)\ \dfrac{\Gamma \rightarrow \Delta, A\ ;\ \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta\ (A \wedge B)}$ $\qquad (\wedge\rightarrow)\ \dfrac{A, B, \Gamma \rightarrow \Delta}{(A \wedge B)\ \Gamma \rightarrow \Delta}$

$(\rightarrow\vee)\ \dfrac{\Gamma \rightarrow \Delta, A, B}{\Gamma \rightarrow \Delta, (A \vee B)}$ $\qquad (\vee\rightarrow)\ \dfrac{A, \Gamma \rightarrow \Delta\ ;\ B, \Gamma \rightarrow \Delta}{(A \vee B)\ \Gamma \rightarrow \Delta}$

$$\dfrac{\dfrac{\dfrac{\overset{ax}{q, r \rightarrow p, q}}{(q \wedge r) \rightarrow (p \vee q)}\ ;\ \dfrac{\dfrac{\dfrac{\overset{ax.}{q, r \rightarrow p, r}}{\neg p, q, r \rightarrow r}}{q, r \rightarrow (\neg p \supset r)}}{(q \wedge r) \rightarrow (\neg p \supset r)}}{\dfrac{(q \wedge r) \rightarrow (p \vee q) \wedge (\neg p \supset r)}{\rightarrow (p \vee q) \wedge (\neg p \supset r), \neg(q \wedge r)}}\ ;\ \dfrac{\dfrac{\overset{ax.}{p \rightarrow p, q}}{p \rightarrow (p \vee q)}\ ;\ \dfrac{\dfrac{\overset{ax.}{p \rightarrow p\ r}}{\neg p, p \rightarrow r}}{p \rightarrow (\neg p \supset r)}}{p \rightarrow (p \vee q) \wedge (\neg p \supset r)}}{\neg(q \wedge r) \supset p \rightarrow (p \vee q) \wedge (\neg p \supset r)}$$

4.) Construct the prenex form of

$$(\forall x\, (Q(x, y) \supset \neg(\neg\exists x Q(x, z) \supset P(z))) \supset \exists x P(x)) .$$

Score: 2 / 2.

$(\forall \underbrace{x}_{x_1} (Q(x, y) \supset \neg(\neg \exists \underbrace{x}_{x_2} Q(x, z) \supset P(z))) \supset \exists \underbrace{x}_{x_3} P(x))$

$(\forall x_1 (Q(x_1, y) \supset \neg(\neg \exists x_2 Q(x_2, z) \supset P(z))) \supset \exists x_3 P(x_3))$

$\exists x_1\, \exists x_2\, \exists x_3\ ((Q(x_1, y) \supset \neg(\neg Q(x_2, z) \supset P(z))) \supset P(x_3))$

5.) Which interpretations are models of the following set of formulae?

$$\{\neg((q \wedge p) \wedge q), (p \equiv (q \vee q))\}$$

$\boxed{2}$ $\varrho(p) = 0, \quad \varrho(q) = 0$

(b) $\varrho(p) = 0, \quad \varrho(q) = 1$

(c) $\varrho(p) = 1, \quad \varrho(q) = 0$

(d) $\varrho(p) = 1, \quad \varrho(q) = 1$

p	q	$\neg((p \wedge q) \wedge q)$	$(p \equiv (q \vee q))$
0	0	1 0 0	1 0
0	1	1 0 0	0 1
1	0	1 0 0	0 0
1	1	0 1 1	1 1

Neptun code: ☐☐☐

6.) Which interpretations are models of the following set of formulae?

$$\{(\neg(q \wedge q) \vee q), ((q \vee p) \equiv p)\}$$

$\boxed{2}$ $\varrho(p) = 0, \quad \varrho(q) = 0$

(b) $\varrho(p) = 0, \quad \varrho(q) = 1$

$\boxed{2}$ $\varrho(p) = 1, \quad \varrho(q) = 0$

$\boxed{2}$ $\varrho(p) = 1, \quad \varrho(q) = 1$

p	q	$(\neg(q \wedge q) \vee q)$	$((q \vee p) \equiv p)$
0	0	1 0 1	0 1
0	1	0 1 1	1 0
1	0	1 0 1	1 1
1	1	0 1 1	1 1

7.) Which interpretations are models of the following set of formulae?

$$\{(q \supset \neg p),\ (((q \supset p) \vee q) \wedge q)\}$$

(a) $\varrho(p) = 0, \quad \varrho(q) = 0$

2 $\varrho(p) = 0, \quad \varrho(q) = 1$

(c) $\varrho(p) = 1, \quad \varrho(q) = 0$

(d) $\varrho(p) = 1, \quad \varrho(q) = 1$

p	q	$(q \supset \neg p)$		$(((q \supset p) \vee q) \wedge q)$		
0	0	1	1	1	1	0
0	1	1	1	0	1	1
1	0	1	0	1	1	0
1	1	0	0			

8.) Which statements are true for the following three formula?

$$A \rightleftharpoons \neg p, \quad B \rightleftharpoons (q \equiv q) \equiv q, \quad C \rightleftharpoons (q \equiv p) \equiv \neg q$$

(a) $A \Leftrightarrow B$,

2 $\{A, B, C\}$ is satisfiable,

2 $A, B \vDash C$,

a) $A \Leftrightarrow B$: $A \vDash B$, $B \vDash A$

$A \nvDash B$ has a model → ~~SAT~~

p	q	$\neg p$	$\neg((q \equiv q) \equiv q)$		
0	0	1	1	1	0
0	1	1	0	1	1
1	0	0	1	1	0
1	1	0	0	1	1

b)

A	B	$(q \equiv p) \equiv \neg q$		
1	0	1	1	1
1	1	0	1	0
0	0	0	0	1
0	1	1	0	0

c)

A	B	$\neg C$
1	0	0
1	1	0
0	0	1
0	1	1

no model → unSAT

9.) How can we formalize the following statement by using a zero-order language with $\{p, q, r\}$?

(r ∧ q) ⊃ ¬p

If it is daytime but there are clouds then the sky is not blue.

$p \leftrightharpoons$ *The sky is blue.* $\quad q \leftrightharpoons$ *There are clouds on the sky.* $\quad r \leftrightharpoons$ *It is daytime.*

(a) $(r \wedge q) \equiv \neg p$

(b) $\neg p \supset (r \wedge q)$

2 $(r \wedge q) \supset \neg p$

(d) $\neg p \equiv \neg r \wedge q$

10.) Let $L^{(1)}$ be a first-order language defined as follows:

$$L^{(1)} = \langle LC, \{x_1, x_2, \ldots\}, \{P, Q, c\}, Term, Form \rangle$$
$$P \in \mathcal{P}(1), \qquad Q \in \mathcal{P}(2), \qquad c \in \mathcal{F}(0).$$

Let $\langle U, \varrho \rangle$ be an interpretation for $L^{(1)}$ such that:

- $U = \{u_1, u_2, u_3\}$
- $\varrho(P) = P' \qquad P'(x) = \begin{cases} 1 & \text{if } x = u_1 \\ 0 & \text{otherwise} \end{cases}$
- $\varrho(Q) = Q' \qquad Q'(x, y) = \begin{cases} 1 & \text{if } x = u_1 \text{ and } y = u_1 \\ 0 & \text{otherwise} \end{cases}$
- $\varrho(c) = u_2$

Let v be an assignment relying on the interpretation such that $v : Var \to U$

$$v(x_1) = u_3, \qquad v(x_2) = u_1$$

Which formulas are true (1) from the ones below?

(a) $\neg\forall x_2 \neg Q(x_2, x_1)$

(b) $\forall x_1 \neg P(x_1)$

2 $\neg\forall x_2 Q(x_2, x_2)$

(d) $\neg\exists x_1 Q(x_1, x_1)$

2 $\exists x_2 \neg P(x_2)$

a) $\neg \forall x_2 \neg Q(x_2, x_1)$
0 1 | 1 $Q(u_1, u_2) = 0$ False
1 $Q(u_1, u_3) = 0$
1 $Q(u_2, u_3) = 0$

b) $\forall x_1 \neg P(x_1)$
0 0 $P(u_1) = 1$ False
1 $P(u_2) = 0$
1 $P(u_3) = 0$

c) $\neg \forall x_2 Q(x_2, x_2)$
1 0 $Q(u_1, u_1) = 1$ True
$Q(u_2, u_2) = 0$
$Q(u_3, u_3) = 0$

d) $\neg \exists x_1 Q(x_1, x_1)$
0 1 $Q(u_1, u_1) = 1$ False
$Q(u_2, u_2) = 0$
$Q(u_3, u_3) = 0$

e) $\exists x_2 \neg P(x_2)$
1 0 $P(u_1) = 1$ True
1 $P(u_2) = 0$
1 $P(u_3) = 0$

23:24 12月18日(月)

121112000044 Chromeで開く 完了

Logic in computer science
Exam

2023. December 11. (Monday) 12:00
INBPA0101E

Result:
93%
26 / 28

121112 000044

Neptun ID: Draft papers:

Name:

The test consists of **10 questions**. The available time to solve the quiz is **100 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

1.) (essay)
2.) (essay)
3.) (essay)

	a	b	c	d	e	f	g	h
4.)	☐	☐	☐	☐	☐	☐	☐	☐
5.)	☐	☐	☐	☐				
6.)	☐	☐	☐	☐				
7.)	☐	☐	☐	☐				
8.)	☐	☐	☐	☐				
	a	b	c	d	e	f	g	h
9.)	☐	☐	☐	☐				
10.)	☐	☐	☐	☐	☐			

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) Define the semantic value of $(A \vee B)$!

Score: 2 / 2.

2.) Define the model of a set of formulae!

Score: 0 / 2.

23:24 12月18日(月)

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2 / 6

3.) Prove the following with sequent calculus.

$$\neg(p \wedge r) \supset q \vDash ((\neg q \supset p) \wedge (q \vee r))$$

Score: 2 / 2.

4.) Select the prenex form of $(\exists x\,(Q(x,y) \supset (\neg\forall x Q(x,z) \wedge P(z))) \supset \forall x P(x))$.

(a) $\exists x \exists u \exists w\,((Q(x,y) \supset (\neg Q(u,z) \wedge P(z))) \supset P(w))$

(b) $\exists x \exists u \forall w\,((Q(x,y) \supset (\neg Q(u,z) \wedge P(z))) \supset P(w))$

(c) $\exists x \forall u \exists w\,((Q(x,y) \supset (\neg Q(u,z) \wedge P(z))) \supset P(w))$

(d) $\exists x \forall u \forall w\,((Q(x,y) \supset (\neg Q(u,z) \wedge P(z))) \supset P(w))$

(e) $\forall x \exists u \exists w\,((Q(x,y) \supset (\neg Q(u,z) \wedge P(z))) \supset P(w))$

(f) $\forall x \exists u \forall w\,((Q(x,y) \supset (\neg Q(u,z) \wedge P(z))) \supset P(w))$

(g) $\forall x \forall u \exists w\,((Q(x,y) \supset (\neg Q(u,z) \wedge P(z))) \supset P(w))$

2 $\forall x \forall u \forall w\,((Q(x,y) \supset (\neg Q(u,z) \wedge P(z))) \supset P(w))$

5.) Which interpretations are models of the following set of formulae?

$$\{((\neg p \wedge q) \equiv q),\ (p \equiv (q \supset q))\}$$

(a) $\varrho(p) = 0, \quad \varrho(q) = 0$

(b) $\varrho(p) = 0, \quad \varrho(q) = 1$

2 $\varrho(p) = 1, \quad \varrho(q) = 0$

(d) $\varrho(p) = 1, \quad \varrho(q) = 1$

Neptun code:

6.) Which interpretations are models of the following set of formulae?

$$\{\, (q \supset (q \supset (q \vee p))),\ \neg(p \equiv q) \,\}$$

(a) $\varrho(p) = 0, \quad \varrho(q) = 0$

$\boxed{2}$ $\varrho(p) = 0, \quad \varrho(q) = 1$

$\boxed{2}$ $\varrho(p) = 1, \quad \varrho(q) = 0$

(d) $\varrho(p) = 1, \quad \varrho(q) = 1$

7.) Which interpretations are models of the following set of formulae?

$$\{\, (p \vee (p \supset q)),\ (q \supset \neg(q \equiv q)) \,\}$$

$\boxed{2}$ $\varrho(p) = 0, \quad \varrho(q) = 0$

(b) $\varrho(p) = 0, \quad \varrho(q) = 1$

$\boxed{2}$ $\varrho(p) = 1, \quad \varrho(q) = 0$

(d) $\varrho(p) = 1, \quad \varrho(q) = 1$

8.) Which statements are true for the following three formula?

$$A \rightleftharpoons q \equiv p, \quad B \rightleftharpoons q \vee \neg q, \quad C \rightleftharpoons q \wedge q \supset \neg p$$

(a) $A \Leftrightarrow B$,

$\boxed{2}$ $\{A, B, C\}$ is satisfiable,

(c) $A, B \vDash C$,

$\boxed{2}$ $(A \vee B)$ is valid.

9.) How can we formalize the following statement by using a zero-order language with $Con = \{p, q, r\}$?

If it is daytime but there are clouds then the sky is not blue.

$p \leftrightharpoons$ *The sky is blue.* $\quad q \leftrightharpoons$ *There are clouds on the sky.* $\quad r \leftrightharpoons$ *It is daytime.*

$\boxed{2}$ $(r \wedge q) \supset \neg p$

(b) $(r \wedge q) \equiv \neg p$

(c) $\neg p \equiv \neg r \wedge q$

(d) $\neg p \supset (r \wedge q)$

10.) Let $L^{(1)}$ be a first-order language defined as follows:

$$L^{(1)} = \langle LC, \{x_1, x_2, \dots\}, \{P, Q, c\}, Term, Form\rangle$$
$$P \in \mathcal{P}(1), \qquad Q \in \mathcal{P}(2), \qquad c \in \mathcal{F}(0).$$

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- $U = \{u_1, u_2, u_3\}$
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- $\varrho(Q) = Q' \qquad Q'(x,y) = \begin{cases} 1 & \text{if } x = u_1 \text{ and } y = u_1 \\ 0 & \text{otherwise} \end{cases}$
- $\varrho(c) = u_2$

Let v be an assignment relying on the interpretation such that $v : Var \to U$

$$v(x_1) = u_3, \qquad v(x_2) = u_1$$

Which formulas are true (1) from the ones below?

2 $\neg\forall x_2 Q(x_2, x_2)$

(b) $\neg\exists x_2 \neg Q(x_2, x_1)$

2 $\exists x_2 \neg P(x_2)$

(d) $\neg\forall x_2 \neg Q(x_2, x_1)$

(e) $\forall x_1 \neg P(x_1)$